GOVERNMENT OF KARNATAKA

GOVERNMENT POLYTECHNIC COLLEGE SIRA

SOFTWARE ENGINEERING AND PRINCIPLES

COURSE CODE:20AI44P

IV SEMESTER

ARTIFICIAL INTELLIGENCE

PREPARED BY:
LAKSHMI G K,
LECTURER/AI

LAB MANUAL

WEEK-1

1. Discuss success and failure stories of software Success story of software:

1. Slack:

Slack is a cloud-based proprietary instant messaging platform. It's a chatroom for the entire company designed to replace email as the primary method of communication. Stewart Butterfield is the CEO and co-founder of Slack. In 2013, Stewart and his team launched Slack, The backend development and development of slack app integrations usually takes close to 360 hours of coding, which amounts to $54,000 on average. Which has transformed business communication. Now slack has net worth of 27.7 billion USD. Slack is used by more than 10million daily active users.

2. Flipkart:

Flipkart is an Indian e-commerce company, was founded in October 2007 by Sachin Bansal and Binny Bansal, Flipkart is the most prominent e-Commerce website in India and is empowering tons of Indian businesses into the competitive online shopping industry. The initial development budget of Flipkart was INR ₹400,000 later it was funded by many funders. In August 2018, U.S. based retail chain Walmart acquired a 77% controlling stake in Flipkart for US$16billion,valuing Flipkart at around $20billion. And now Flipkart has more than 350million customers and offering 150 million products across 80+ categories with net worth of 40 billion

3. Phone pe:

Phone pe is a mobile payment platform. Phone pe wasfoundedinDecember2015 Its founder are Sameer nigam, Rahul Chari. Phone pe investment was 12 million. It went live in august 2016.It could be linked to a single debit/credit card or bank account .This app is available in 11Indian languages. It helps to on line transaction easily with safe from cyber criminals. The phone pe was based on UPI(unified payment interface).Phone pe was secured by UPI password. It is very easy to recharge your mobile number through phone pe. One of the other features of the phone pe app is the option to send and receive money using a QR code.

Failure Story of software:

1. Software Fault in Airbus A400M plane:

In May 2015 The Airbus A400M crashed near Seville ,on 9May after a failed emergency landing during its first flight. It was a fatal crash that happened in Spain. Data needed to run the engines had been accidentally erased and lead to crash and cost the loss of 23 billion and 4 crew members

2. Ransom ware attack to colonial pipeline:

In 2021 hackers get access to computer system and essentially hold a company hostage for money. The attack began when a hacker group identified as Dark Side accessed the Colonial Pipeline network. The attackers stole 100 gigabytes of data within a two-hour window. Following the data theft, the attackers infected the Colonial Pipeline IT network with ransom ware that affected many computer systems , including billing and accounting.

One of the largest fuel pipeline in the US, colonial pipeline was hit by a cyberattack which was made colonial CEO to pay 4.4 million dollars to get company's network up and running again.

1. Nest thermostats leaves:

Nest thermostat is a home app or nest app .Inmid -Jan2016 nest smart thermostat was developed. It was owned by Google. It cost was $50 million. This software is used to heat the homes in the cold time. While it's actively heating or cooling, the error in this software was forcing the device batteries to drain, leaving it unable to control temperature, so customers were unable to heat their homes.

Software ethical practices:

Profession:

Software engineers shall advance the integrity and reputation of the profession consistent with the public interest. In particular, software engineers shall, as appropriate:

- Help develop an organizational environment favorable to acting ethically.
- Promote public knowledge of software engineering.
- Extend software engineering knowledge by appropriate participation in professional organizations, meetings and publications.
- Support, as members of a profession, other software engineers striving to follow this Code.
- Not promote their own interest at the expense of the profession, client or employer.
- Avoid as sociations with businesses and organizations which are in conflict with this code.
- Take responsibility for detecting, correcting, and reporting errors in software and associated documents on which they work.

Laws for IT industry in India:

The Computer Misuse Act 1990

The Computer Misuse Act creates three new offences that can briefly be described as:

- Unauthorized access to a computer;
- Unauthorized access to a computer with intention to commit a serious crime; and
- Unauthorized modification of the content of a computer

Section1:of the Computer MisuseAct1990 states that, a person is guilty of an offence if

- Person causes a computer to perform any function with intent to secure access to any program or data held in any computer
- Person access he intends to secure is unauthorized and
- Person knows at the time when he caused the computer to perform the function that is the case.

Section2:of the Act is concerned with gaining unauthorized access to a computer with the intention of committing a more serious offence.

- A blackmailer might attempt to gain unauthorized access to medical records
- A terrorist might try to get access to a computer system to cause accidents to happen

Section3:of the Act states that: a person is guilty of an offence if

- He does any act which causes an unauthorized modification of the contents of any computer.
- At the time when he does the act he has the requisite intent and there is knowledge

The Data Protection Act 1998 (DPA)

To determine the appropriateness of security measures ,you should consider all of the following:

- Unauthorized or unlawful processing of personal data.
- Accident all loss or destruction of, or damage to ,personal data
- Implementation costs
- Technological developments
- The nature of the data -sensitive personal data will merit particular attention
- The harm that might result from unauthorized or unlawful processing or from accidental loss destruction and damage to the data